GREEN'S vs STOKES' THEOREM

Green's circulation theorem is Stokes on a plane.

GREEN'S THEOREM

Planar region D in the xy-plane
[R Green Plot Here]

$$\oint_C (P\,\mathrm{d}x + Q\,\mathrm{d}y) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) \mathrm{d}A$$

Scalar curl integrated over a planar region.

STOKES' THEOREM

Surface S bounded by closed curve C
[R Stokes Plot Here]

$$\oint_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{r} = \iint_S (\nabla \times \boldsymbol{F}) \cdot \mathrm{d}\boldsymbol{S}$$

Curl flux over a surface.

Recover Green from Stokes

$$\boldsymbol{F} = (P, Q, 0) \Big| \widehat{\boldsymbol{n}} = (0, 0, 1) \Big| \boldsymbol{F} \cdot \mathrm{d}r = P\mathrm{d}x + Q\mathrm{d}y \Big| (\nabla \times \boldsymbol{F}) \cdot \widehat{\boldsymbol{n}} \Big| \mathrm{d}S = \mathrm{d}A$$

On a flat xy-surface, Stokes reduces exactly to Green's circulation form.

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